



GPS L5C SDR



University of Colorado
Boulder



北京理工大学
BEIJING INSTITUTE OF TECHNOLOGY

Introduction (1/7)

GPS signal parameters

Signals		Code Freq.	Code Len.	Nav. Data		Modulation	Freq.	Band Width	Min.
		[Mcps]	[chip]	Type	D(t)/d(t)[bps]		[MHz]	[MHz]	Received Power[dBW]
L1 C/A		1.023	1023	NAV	50/50	BPSK-R(1)	1572.42	2.046	-158.5
L1 P(Y)		10.23	7 days	NAV	50/50	BPSK-R(10)		20.46	-161.5
L1 M		5.115	Unknown	MNAV	Unknown	BOC(10,5)		30.69	Unknown
L1C	L1 CD	1.023	10230	CNAV-2	50/100	BOC(1,1)		4.092	-157.0
	L1 CP	1.023	10230×1800	Pilot Channel		TMBOC(6,1,4/33)		14.332	
L2 P(Y)		10.23	7 days	NAV	50/50	BPSK-R(1)	1227.6	20.46	-161.5
L2C	CM	0.5115×2	10230	CNAV	25/50	BPSK-R(1)		2.046	-160.0
	CL		767250	Pilot Channel					
L2M		5.115	Unknown	MNAV	Unknown	BOC(10,5)		30.69	Unknown
L5C	L5I	10.23	10230×10	CNAV	50/100	QPSK-R(10)	1176.45	20.46	-157.9 for each component
	L5Q	10.23	10230×20	Pilot Channel				20.46	

Introduction (2/7)

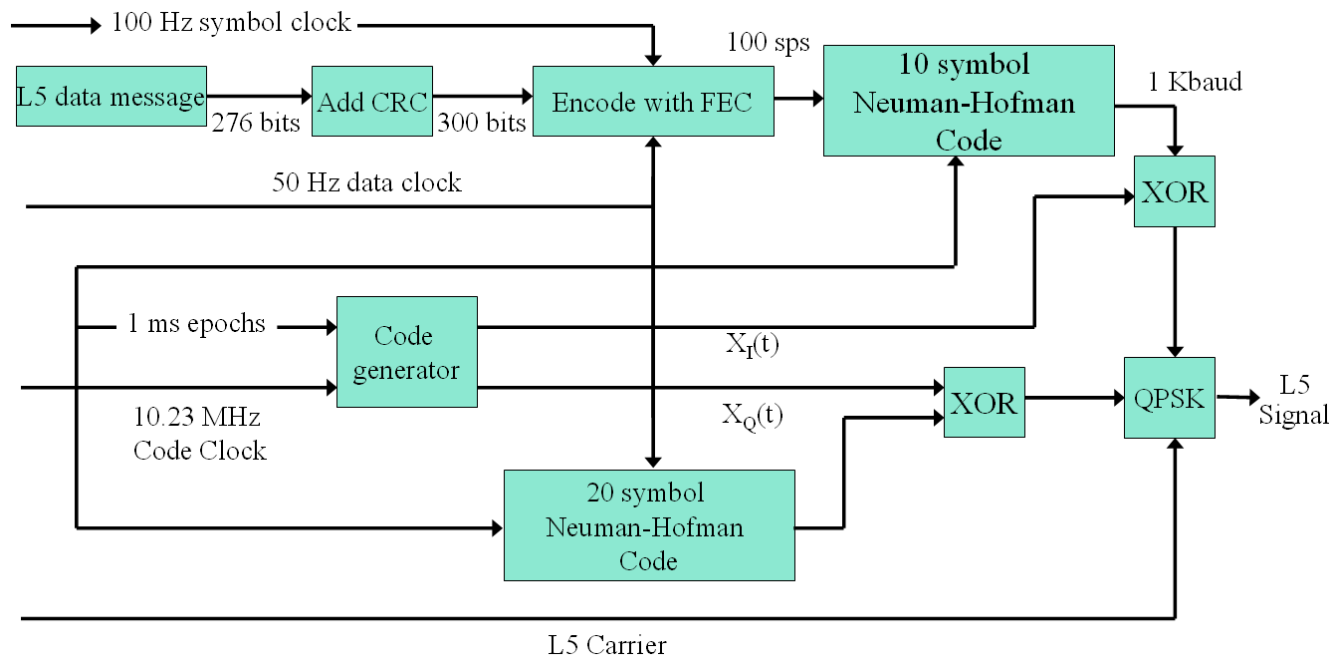
List of SVs transmitting L5 signals (*)

Satellite	Launch (UTC)	Launch site	Block	Nº	SVN	PRN	Slot
USA-213	5/28/2010	CCAFS SLC-37B	IIF	1	62	25	B2
USA-232	7/16/2011	CCAFS SLC-37B	IIF	2	63	1	D2
USA-239	10/4/2012	CCAFS SLC-37B	IIF	3	65	24	A1
USA-242	5/15/2013	CCAFS SLC-41	IIF	4	66	27	C2
USA-248	2/21/2014	CCAFS SLC-37B	IIF	5	64	30	A3
USA-251	5/17/2014	CCAFS SLC-37B	IIF	6	67	6	D4
USA-256	8/2/2014	CCAFS SLC-41	IIF	7	68	9	F3
USA-258	10/29/2014	CCAFS SLC-41	IIF	8	69	3	E1
USA-260	3/25/2015	CCAFS SLC-37B	IIF	9	71	26	B5
USA-262	7/15/2015	CCAFS SLC-41	IIF	10	72	8	C5
USA-265	10/31/2015	CCAFS SLC-41	IIF	11	73	10	E6
USA-266	2/5/2016	CCAFS SLC-41	IIF	12	70	32	F5

(*) Wikipedia: https://en.wikipedia.org/wiki/List_of_GPS_satellites

Introduction (3/7)

Modulation scheme for the GPS L5 Signals



Two PRN ranging codes :

- ◆ **L5I-code (data channel) and L5Q-code (pilot channel)**
- ◆ **I5-code is modulated with a 10-bit Neuman-Hofman code**
- ◆ **Q5-code is modulated with a 20-bit Neuman-Hofman code**

Introduction (4/7)

L5C signal characteristics:

- ✦ Civilian, safety of life signal
- ✦ Improves signal structure for enhanced performance
- ✦ Higher transmitted power than L1/L2 signal ($\sim 3\text{db}$)
- ✦ Wider bandwidth provides a 10x processing gain
- ✦ Longer spreading codes (10x longer than C/A)
- ✦ Uses the Aeronautical Radio navigation Services (ARNS) band

Introduction (5/7)

Convolution Encoder of CNAV message in L5I

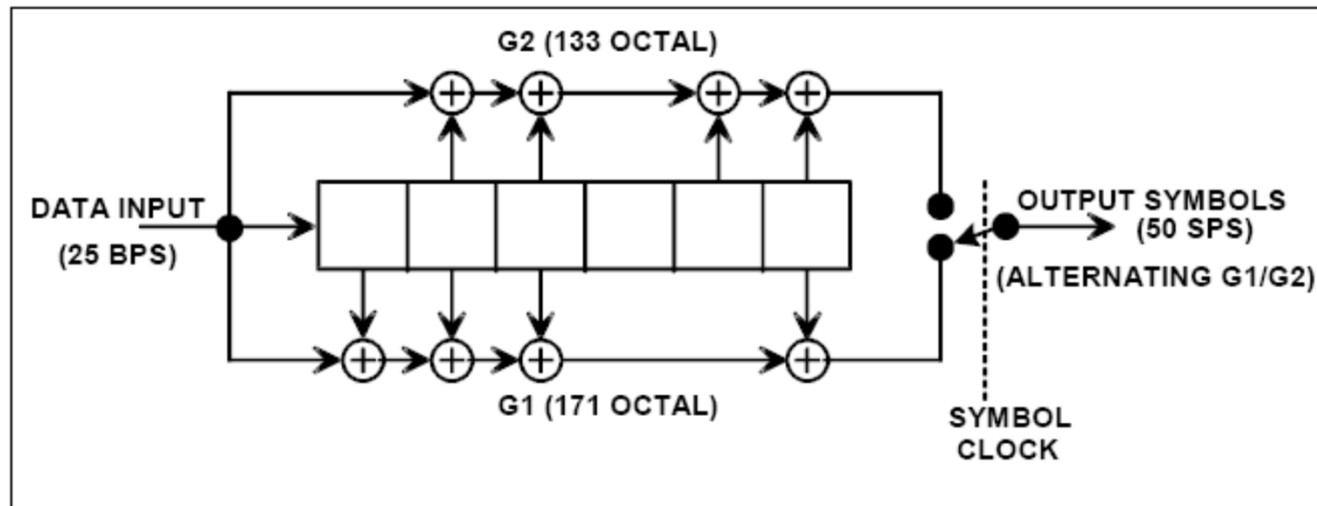


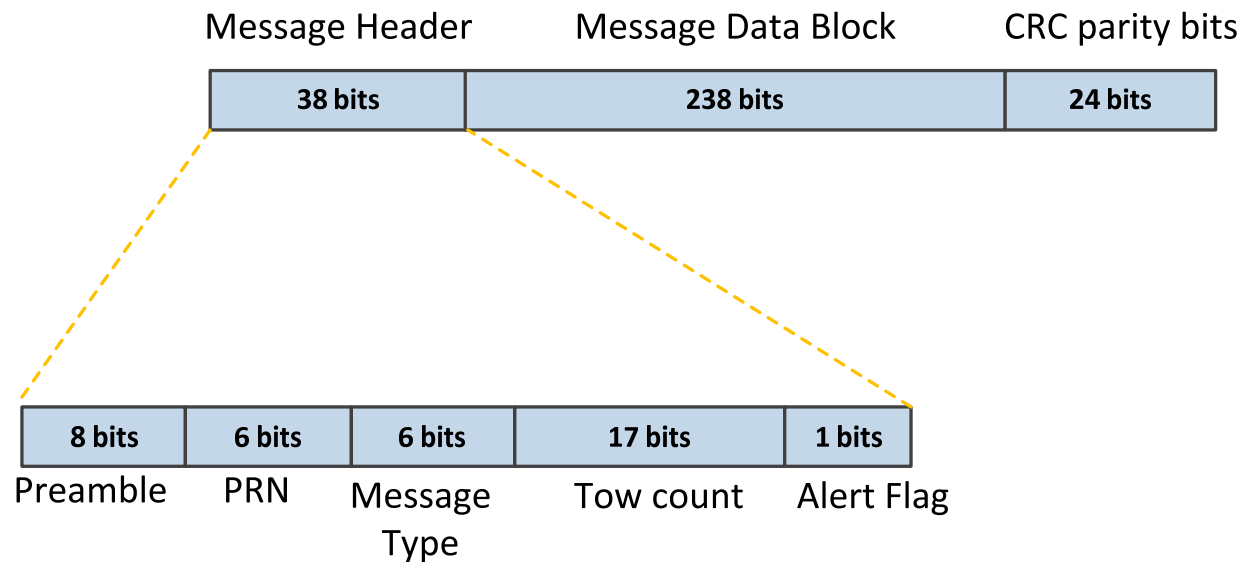
Fig. 3-7 of IS-GPS-200D

- ◆ ½ fixed convolutional code + CRC-24Q
- ◆ **Viterbi Decoder** / Fano Decoder
- ◆ Same as Galileo I/NAV

Introduction (6/7)

Message structure of CNAV in L5I signals

300 bits long CNAV message



- ◆ Same as L2C CNAV
- ◆ Forward error correction (FEC) encoded by 1/2 convolutional code.
- ◆ Transmit a complete ephemeris every **24** seconds.

Introduction (7/7)

CNAV Message contents

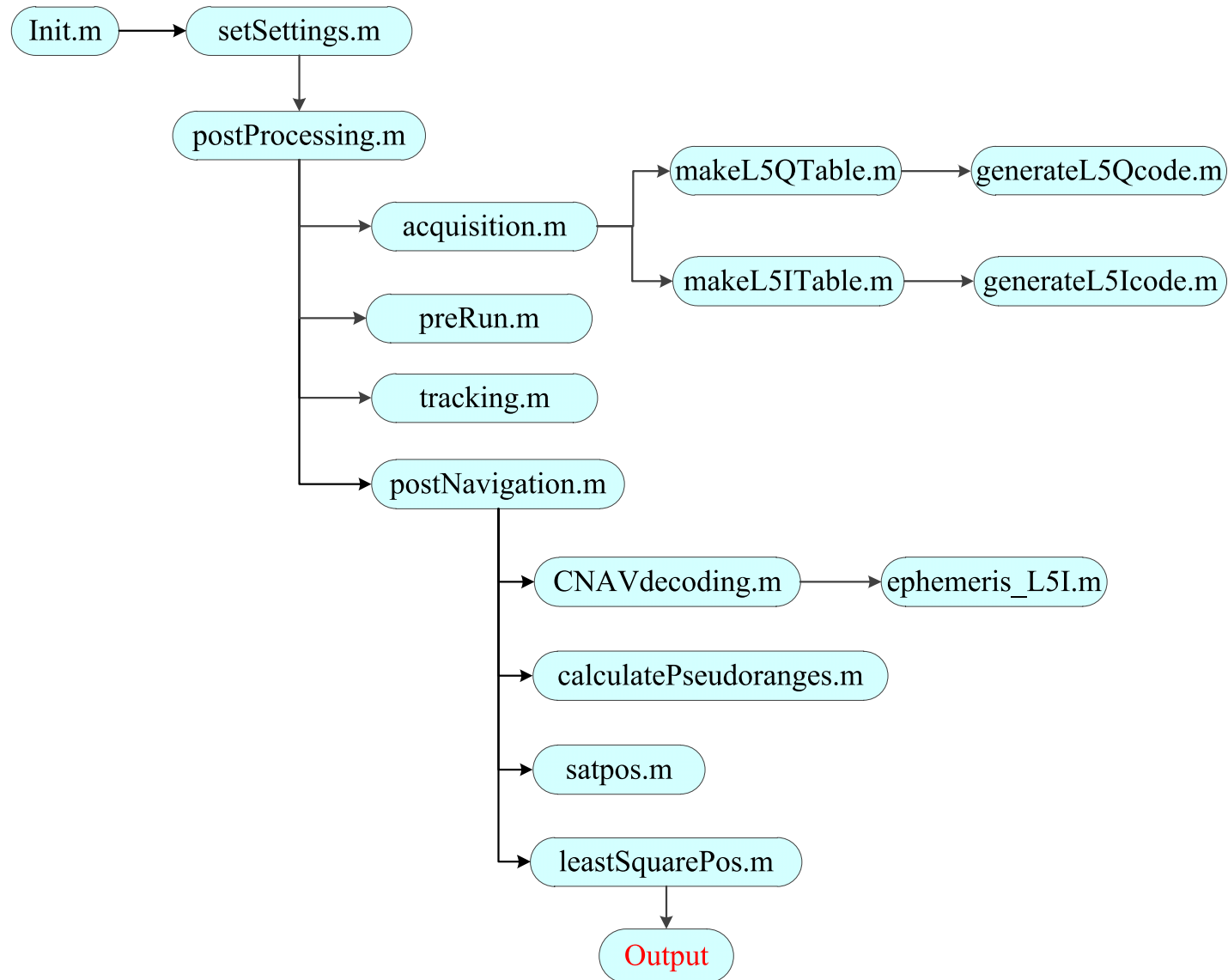
➤ **Requisite for receiver fix:**

- **Type 10**
- **Type 11**
- **Any one of 30-37**

➤ **Broadcast Interval is half of that in L2C**

Type No.	Message Content	Max. Broadcast Intervals
10	Ephemeris	24 seconds
11	Ephemeris	24 seconds
30-37	Clock	24 seconds
30	Inter-Signal Correction and Ionospheric Data	144 seconds
33	UTC	144 seconds
35	GPS/GNSS Time Offset	144 seconds
31 or 12	Reduced Almanac	10 minutes
34 or 13	Differential Correction	15 minutes
32	Earth Orientation Parameters	15 minutes
14	Differential Correction	15 minutes
37	Midi Almanac	1 hours
36 or 15	Text Message	Not determined

SDR processing procedure



initSettings.m (1/2)

➤ Acquisition settings

- Single sideband for carrier freq. search
 - `settings.acqSearchBand = 5000;`
- Carrier freq. search step
 - `settings.acqStep = 500;`
- Sampling rate threshold for downsampling
 - If input signal sampling freq. is too high for FFT computation, it can be resampled to a lower freq. to save processing time.
 - `settings.resamplingThreshold = 50e6;`
- Enable/disable use of downsampling for acquisition
 - `settings.resamplingflag = 1;      % 0 – Off, 1 - On`

initSettings.m (2/2)

➤ Code parameter setting

- L5 PRN code rate is 10.23Mcps, and the code length is 10230 chips.
 - `settings.codeLength = 10230;`
 - `settings.codeFreqBasis = 10.23e6;`

➤ Integration time for DLL and PLL

- `settings.intTime = 0.001; %[s]`

➤ Tracking loops settings

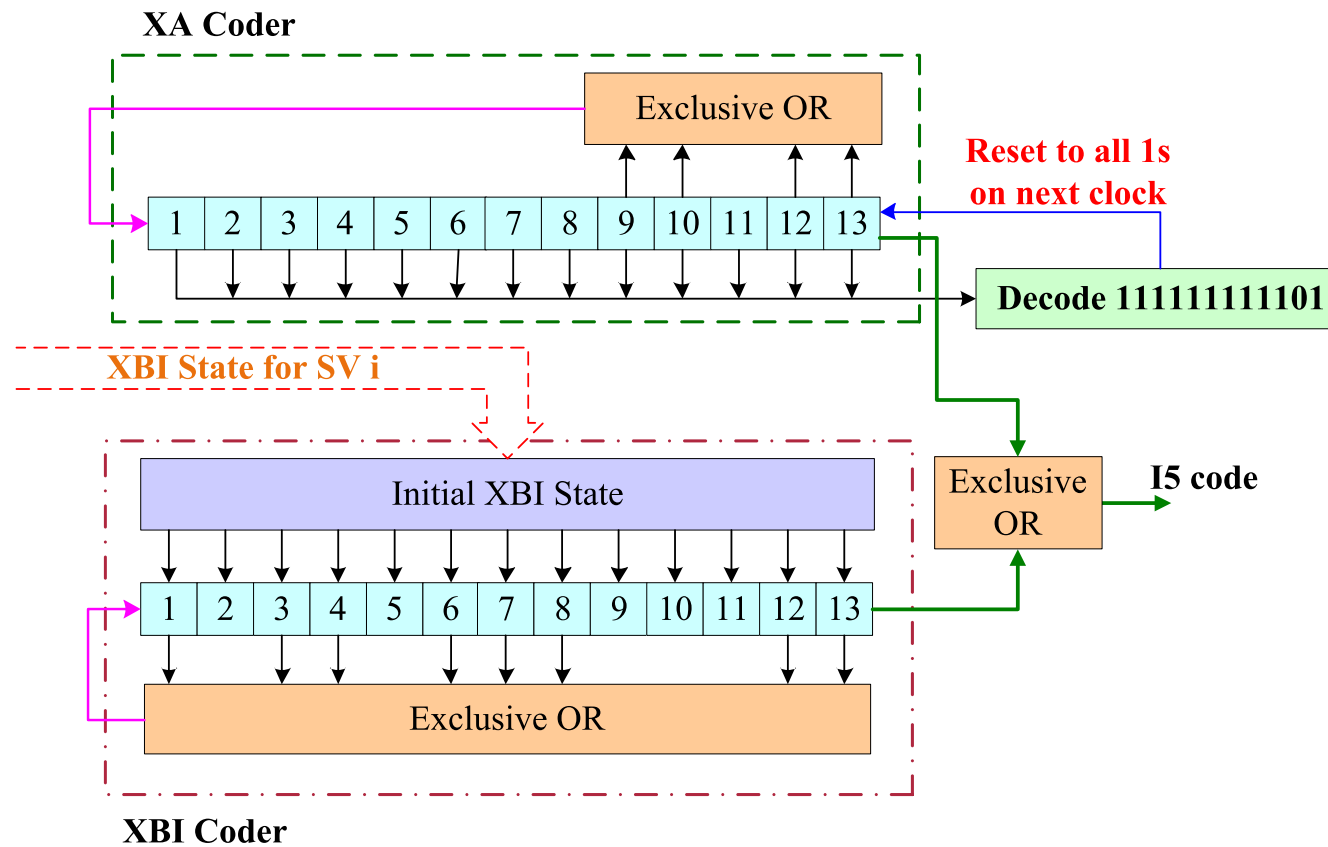
- Enable/disable use of **pilot channel** for tracking.
 - `settings.pilotTRKflag = 1; % 0 – Off 1 - On`

➤ L5 carrier frequency

- `settings.carrFreqBasis = 1176.45e6;`

generateL5Icode.m (1/4)

Generation of I5 code



generateL5Icode.m (2/4)

➤ Generating XA code

- (1) Set Feedback position of exclusive OR

- `xa_FeedbackPos = [9 10 12 13];`

- (2) Initialize XA register and XA codes output

- `xa_reg = ones(1,13) * -1;`

- `XA = zeros(1,CodeLength);`

- (3) Set reset state

- `reset_state = [-1,-1,-1,-1,-1,-1,-1,-1,-1,-1,-1,1,-1];`

- (4) Generate XA output

- `for ind = 1:CodeLength`

- `XA(ind) = xa_reg(end);`

- `if isequal(xa_reg,reset_state)`

- `xa_reg = ones(1,13) * -1;`

- `else`

- `...`

- `end`

- `end`

generateL5Icode.m (3/4)

➤ Generating XBI code

- (1) Initial-state table (See “IS-GPS-200D” for more details)

- `l5i_init = [...];`

- (2) Feedback position of exclusive OR operation

- `xbi_FeedbackPos = [1 3 4 6 7 8 12 13];`

- (3) Initialize XBI register and XBI codes output

- `xbi_reg = ones(1,13) * -1;`

- `XBI = zeros(1,CodeLength);`

- (4) Produce XBI initial states

- `for ind = 1:resetPos`

- `feedback = prod(xbi_reg(xbi_FeedbackPos));`

- `xbi_reg = circshift(xbi_reg',1)';`

- `xbi_reg(1) = feedback;`

- `End`

generateL5Icode.m (4/4)

➤ Generating XBI code

- (5) XBI output for ind = 1:CodeLength
 - for ind = 1:CodeLength
 - XBI(ind) = xbi_reg(end);
 - feedback = prod(xbi_reg(xbi_FeedbackPos));
 - xbi_reg = circshift(xbi_reg',1)';
 - xbi_reg(1) = feedback;
 - end
- (6) Produce I5 PRN code output
 - L5Icode = XBI .* XA;

generateL5Qcode.m (1/1)

➤ Difference from “generateL5Icode.m”

- Initial-state table (See “IS-GPS-200D” for more details)

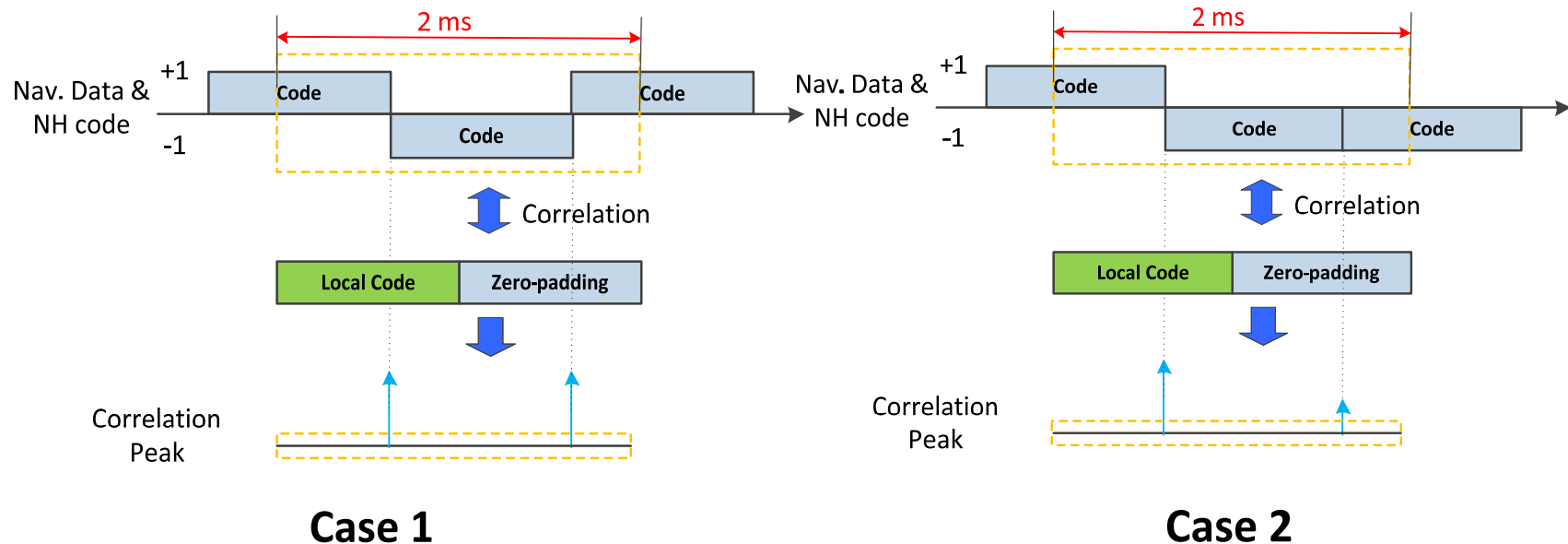
- `I5Q_init = [...];`

- Other parts are same as “generateL5Icode.m”

acquisition.m (1/6)

➤ Acquisition Method: coarse acquisition

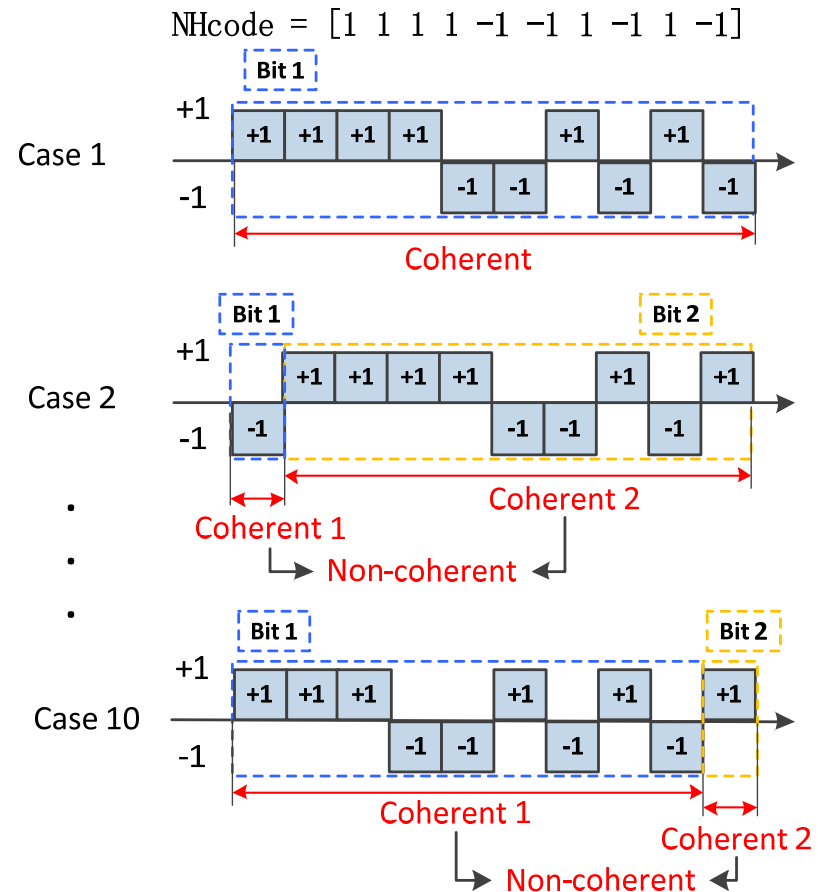
- 1 ms coherent integration + 1ms **Zero-padding FFT**
- Circular correlation
- Integrated power from processing both I5 and Q5 codes **are summed** to improve sensitivity



acquisition.m (2/6)

➤ Acquisition Method: fine acquisition

- Serial search strategy for fine carrier frequency using L5Q pilot signals
- $[-250, +250]$ uncertainty with 25 Hz search step
- 20 cases of NH code combination
- Coherent combination



acquisition.m (3/6)

➤ Acquisition implementation

- Selectable down-sampling is same as that in L2C SDR
- Coarse acquisition: processing power of I5 and Q5 codes are summed:

```
. L5ICodesTable = makeL5ITable(PRN,settings);  
. L5QCodesTable = makeL5QTable(PRN,settings);  
. localL5ICode = [L5ICodesTable, zeros(1,samplesPerCode)];  
. localL5QCode = [L5QCodesTable, zeros(1,samplesPerCode)];  
. L5ICodeFreqDom = conj(fft(localL5ICode));  
. L5QCodeFreqDom = conj(fft(localL5QCode));  
. ...  
. convCodeIQ1 = IQfreqDom1 .* L5ICodeFreqDom;  
. convCodeIQ2 = IQfreqDom1 .* L5QCodeFreqDom;  
. results(frqBinIndex, :) = abs(ifft(convCodeIQ1)) + abs(ifft(convCodeIQ2));
```

acquisition.m (4/6)

➤ Acquisition implementation: fine acquisition

- Set related parameters and initialize arrays

```
· NHcode = [1 1 1 1 -1 -1 1 -1 1 -1];  
· FineFrqBins = zeros(1, NumOfFineBins);  
· ...
```

- Search different frequency bins

```
· for FineBinIndex = 1 : NumOfFineBins  
·     FineFrqBins(FineBinIndex) = frqBins(frequencyBinIndex) -  
·                                     250 + 25 * (FineBinIndex - 1);  
· ...  
· End
```

- Search different NH code combinations

```
· for comIndex = 1:10  
·     % Wipe off NH code  
·     sumPerCodeNH = sumPerCode .* NHcode;  
·     % Maximal coherent power for different NH code combinations  
·     maxPower = max(maxPower,abs(sum(sumPerCodeNH)));  
·     ...  
· end
```

acquisition.m (5/6)

➤ Acquisition implementation: fine acquisition

- Find the fine carrier freq. corresponding to the largest coherent power

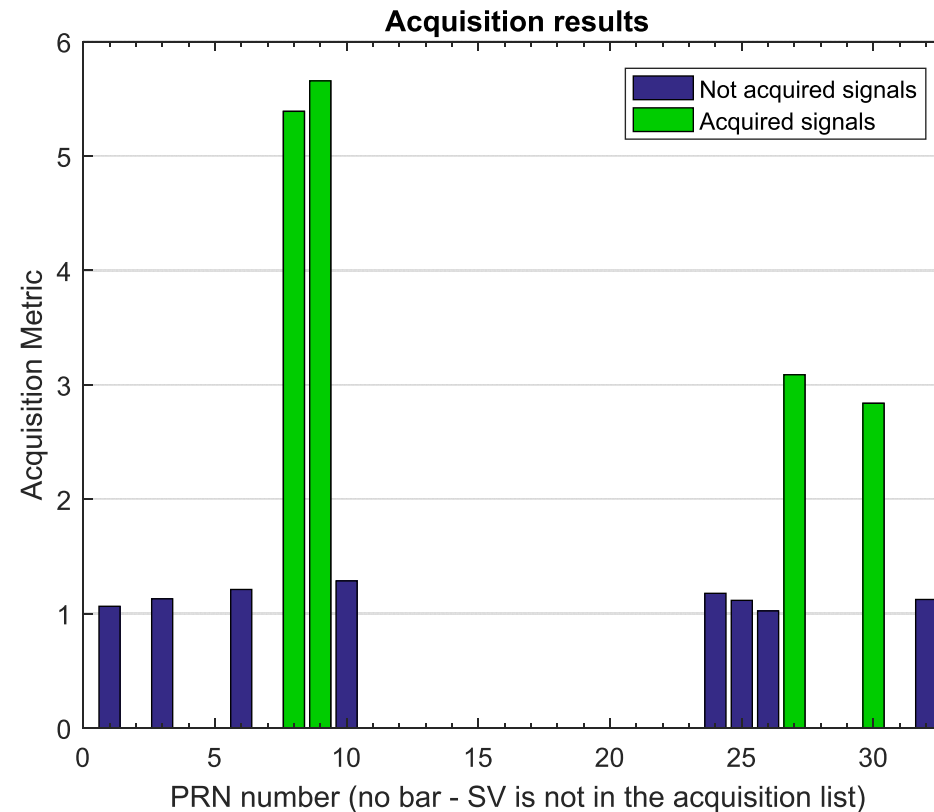
- `[~,maxFinBin] = max(FineResult);`
- `acqResults.carrFreq(PRN) = FineFrqBins(maxFinBin);`

- Change fine acquisition results to correspond with original sampling frequency, if down-sampling is carried out

- The same as that in L2C SDR

acquisition.m (6/6)

➤ Acquisition Results (*)



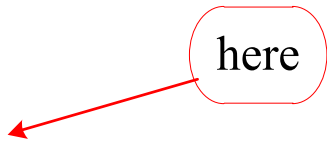
(*) `settings.filename = 'GPS_L5C_IF_signal.bin';`

preRun.m (1/1)

- L5 code tracking is more sensitive to code frequency errors than that of L1 C/A, due to:
 - L5 code chip is 10 times shorter than L1 C/A
 - Nominal L5 code freq. cannot be directly used to initialize tracking
- Use code freq. converted from acquired carrier freq.
 - `channel.codeFreq = 0; % Used as the center frequency of the code NCO`
 - **One problem:** the computed carrier Doppler is opposite to its true value (perhaps related to front-end design)

```
· channel(ii).codeFreq = settings.codeFreqBasis - ...  
· (channel(ii).acquiredFreq - settings.IF)/...  
· settings.carrFreqBasis * settings.codeFreqBasis;
```

here



tracking.m (1/3)

➤ Combined tracking of data (L5I) and pilot (L5Q)

- Combined code-tracking error estimate D

$$D = \frac{D_d + D_p}{2}$$

$$D_d = \frac{\sqrt{I_{E,d}^2 + Q_{E,d}^2} - \sqrt{I_{L,d}^2 + Q_{L,d}^2}}{\sqrt{I_{E,d}^2 + Q_{E,d}^2} + \sqrt{I_{L,d}^2 + Q_{L,d}^2}} \quad D_p = \frac{\sqrt{I_{E,p}^2 + Q_{E,p}^2} - \sqrt{I_{L,p}^2 + Q_{L,p}^2}}{\sqrt{I_{E,p}^2 + Q_{E,p}^2} + \sqrt{I_{L,p}^2 + Q_{L,p}^2}}$$

where, D_d and D_p are code tracking errors estimated from data and pilot signals, respectively.

- Combined carrier-tracking error estimate φ

$$\varphi = \frac{\varphi_d + \varphi_p}{2} \quad \varphi_d = \tan^{-1} \left(\frac{Q_d}{I_d} \right)$$

$$IQ_p = (I_p + j \cdot Q_p)e^{-j\pi/2} \quad \varphi_p = \tan^{-1} \left(\frac{\text{imag}(IQ_p)}{\text{real}(IQ_p)} \right)$$

where, φ_d and φ_p are carrier tracking errors estimated from data and pilot signals, respectively.

As pilot carrier phase is $\pi/2$ rad ahead of the data carrier phase, here we rotate the pilot channel correlation value $I_p + j \cdot Q_p$ by $\pi/2$ back to that of the data channel.

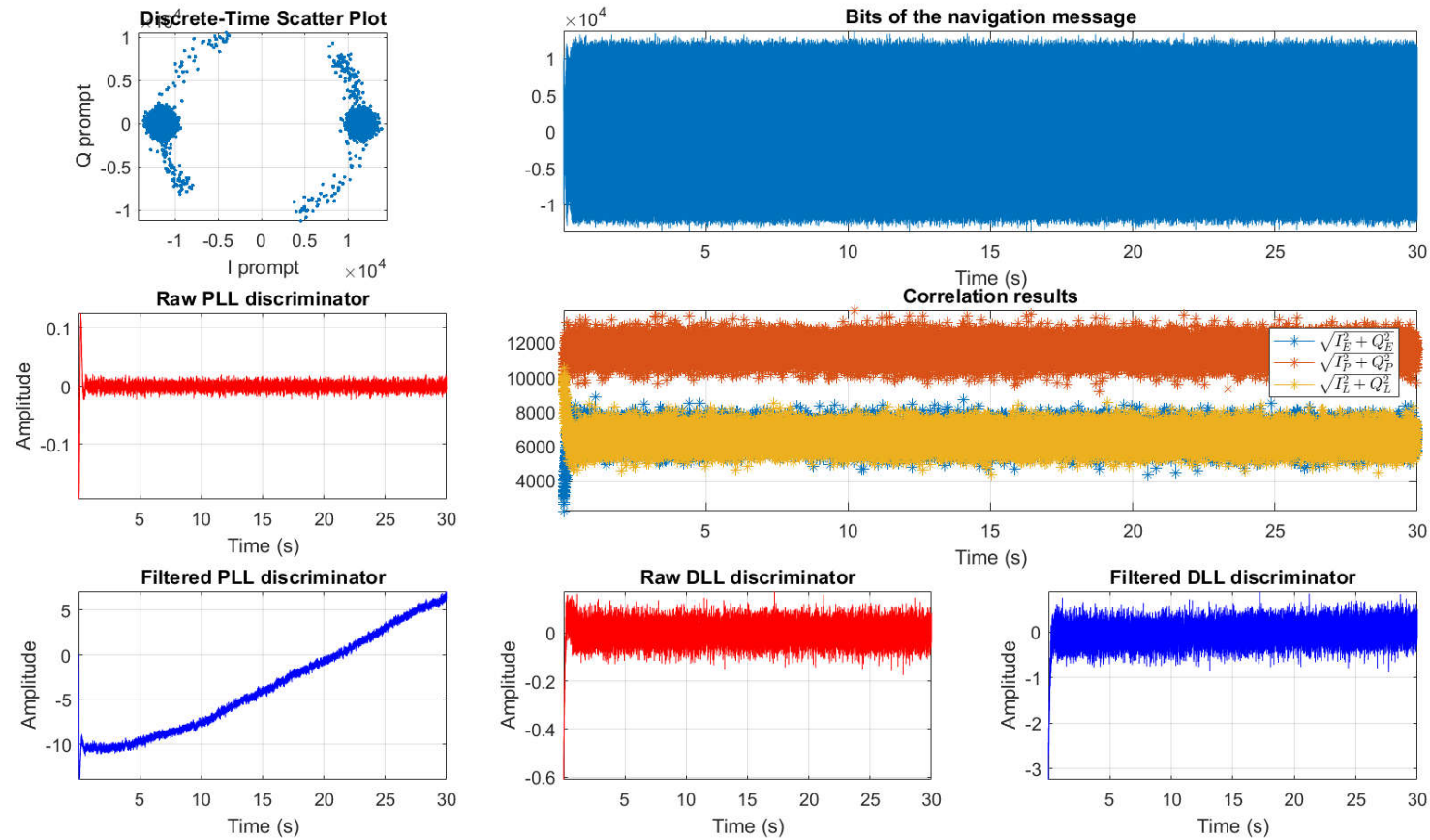
tracking.m (2/3)

➤ Other difference from L2C SDR

- Use code freq. converted from carrier freq. to initialize code NCO
 - `codeFreq = channel(channelNr).codeFreq;`

tracking.m (3/3)

➤ Data channel tracking result of PRN# 9:



postNavigation.m (1/1)

➤ Main changes compared with the L2C SDR

- CNAV data and ephemeris decoding
 - See “CNAVdecoding.m” and “ephemeris.m”
- Calculation of signal transmitting time
 - See “calculatePseudoranges.m” (same as L2c SDR)
- Calculation of pseudorange and rate
 - See “calculatePseudoranges.m” (same as L2c SDR)
- Position computation for SVs
 - See “satpos.m”

CNAVdecoding.m (1/3)

➤ Step 1: Bit synchronization

- (1) Set search offset to avoid noise due to tracking loop transients

- `searchStartOffset = 800;`

- (2) Set parameters and allocate variable space

- `NHcode = [1 1 1 1 -1 -1 1 -1 1 -1];`

- `bitSyncPos = inf(1, settings.numberOfChannels);`

- (3) Correlation between NH codes and tracking inphase power

- ...

- `tlmXcorrResult = xcorr(I_P_Input, NHcode);`

- ...

- (4) Find the start position of the first bit

- `for ind = 1: fix(length(tlmXcorrResult)/2)`

- `navBits = abs(tlmXcorrResult(ind:10:end-10));`

- `if all((abs(navBits-10)<0.01),2)`

- `bitSyncPos(channelNr) = ind + searchStartOffset;`

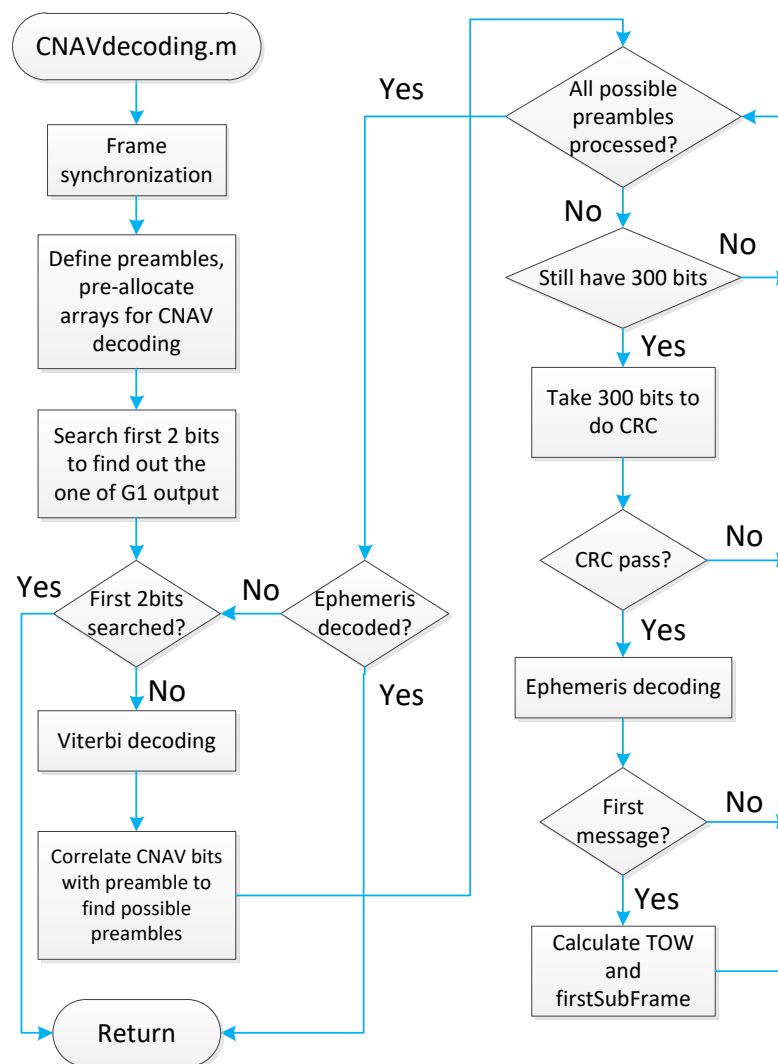
- `break`

- `end`

- `end`

CNAVdecoding.m (2/3)

➤ Step 2: Frame synchronization and CNAV data decoding



Implementation procedure
(similar to L2C SDR)

CNAVdecoding.m (3/3)

➤ Step 2: Main changes compared with L2C SDR

- - Wipe off NH code
- `NHcode = [1 1 1 1 -1 -1 1 -1 1 -1];`
- ...
- `dataBits = sum(I_P_group .* repmat(NHcode,bitsNum,1),2)';`
- - Start position of first subframe (each bit lasts 10ms)
- `firstSubFrame = (index(i) * 2 + G1orG2 - 2)*10 + (bitSyncPos-1);`

ephemeris.m (1/1)

➤ Difference from L2C SDR

- The group delay differential correction terms.

- `eph.ISC_L5I = twosComp2dec(navBitsBin(167:179)) * 2^(-35);`

- Each message lasts 6s rather than 12s in L2C

- `TOW = bin2dec(navBitsBin(21:37)) * 6 - 6;`

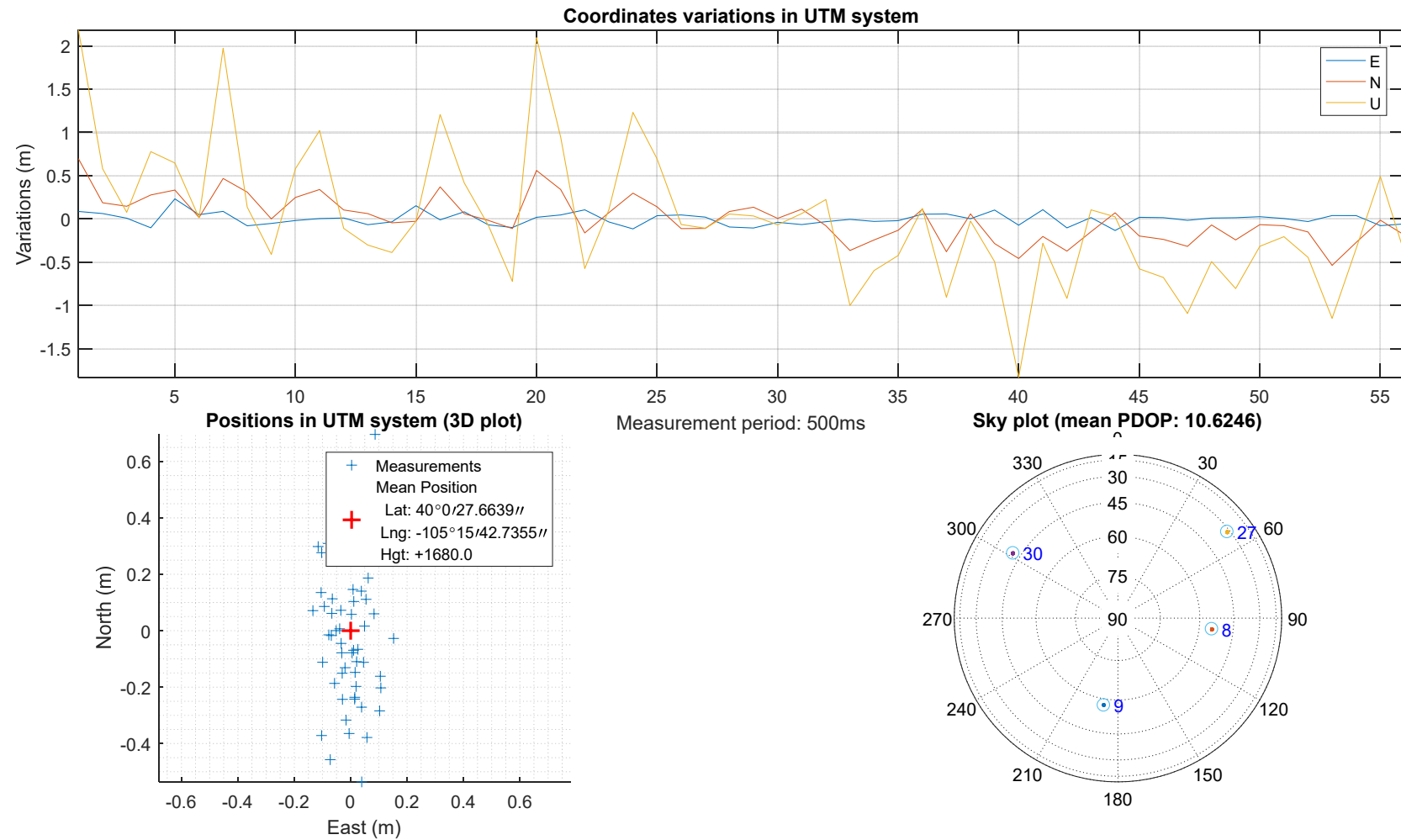
satpos.m (1/1)

➤ Difference from L2C SDR

- Group delay for clock correction

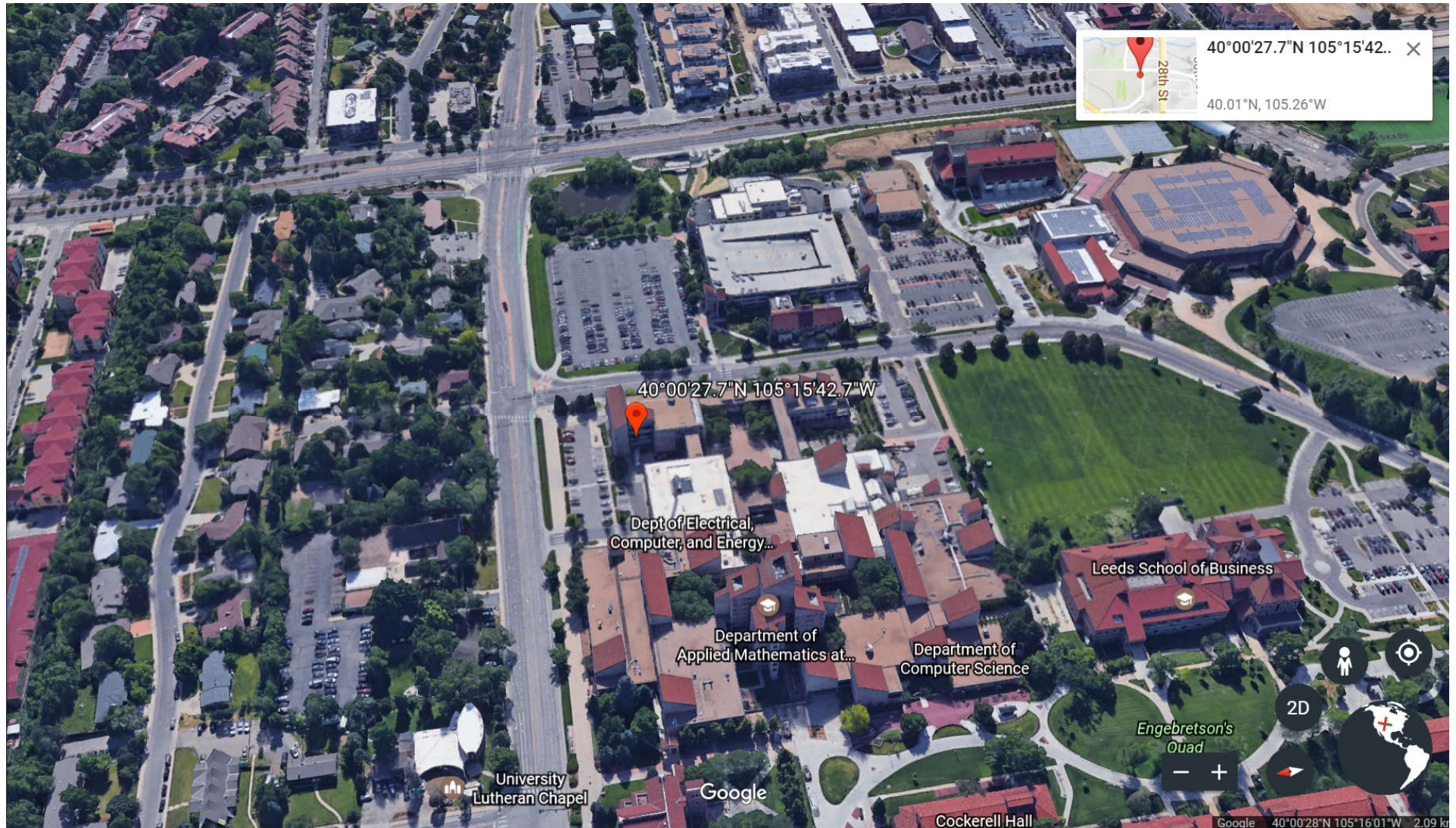
```
.   if(eph(prn).idValid(3) == 30)
.       satClkCorr(satNr) = satClkCorr(satNr) - ...
.           eph(prn).T_GD + eph(prn).ISC_L5I;
.   end
```

Fix results (1/2)



*** The IF signal is collected on the roof of the Discovery Learning Center, CU-Boulder, USA.**

Fix results (2/2)



References

- **IS-GPS-705D, Navstar GPS Space Segment/Navigation User Interfaces. Sep. 24, 2013.**
- **Xie Gang. Principles of GNSS: GPS, Glonass and Galileo. Publishing house of electronics industry, Beijing, Sep. , 2013.**
- **Kai, B., Akos, D. M., Bertelsen, N., Rinder, P., & Jensen, S. H.. A Software-Defined GPS and Galileo Receiver. Birkhäuser Boston, 2007.**



Thank you!

Yafeng Li lyf8118@126.com



University of Colorado
Boulder



北京理工大学
BEIJING INSTITUTE OF TECHNOLOGY

